

Lyra EOD Wednesday — 2026-06-03

Date: 2026-06-03 (Wednesday) Time: ~20:15 EDT Status: EOD — Casey signal

1. Wednesday Substantive Arc

Headline: Wednesday morning closed at 9-of-12 K3 v0.16 verification gates substantively addressed via THREE-mechanism substrate framework (chirality projection + Weyl branching + Lorentz integration); Wednesday afternoon produced three new Lyra framework documents (Mehler + m_e/m_P + SSG-15) with extensive multi-CI bidirectional discipline cascade producing sharpened substrate-mechanism content through correction process.

2. Lyra Wednesday Output Inventory

14 substantive items:

Wednesday morning (5 registry addenda v0.9 → v0.13): 1. SSG Registry v0.9: Elie Toy 3738 $SO(5) \rightarrow SO(3,1)$ Weyl branching + SSG-11 $V_{(2,0)}$ graviton-sector physical content identified 2. SSG Registry v0.10: Cal #194 PASS + Elie 3739 $V_{(3/2,1/2)}$ gen-2 candidate + DUAL ROLE + Casey #14 WAIT 3. SSG Registry v0.11: K-type $\neq$ mass mechanism distinction per Elie 3740 + Keeper K3 v0.15 4. SSG Registry v0.12: THREE-Mechanism Substrate Framework consolidated (chirality + Weyl + Lorentz integration) 5. SSG Registry v0.13: 3-gen spinor-tower row COMPLETE per Elie 3742 $V_{(5/2, 1/2)}$ + K3 v0.16 12-gate framework

Wednesday afternoon (3 new framework documents): 6. notes/Lyra_Mehler_Matrix_Element_Framework_v0_1.md (~10:35 EDT): operator-level mass mechanism via $M_{op} = \sqrt{H_B}$ + Lorentz integration; Casimir eigenvalues spinor-tower computed; 4-of-12 K3 v0.16 gates at FRAMEWORK PRE-STAGE 7. notes/Lyra_me_mP_1156_Correction_Framework_v0_1.md (~10:45 EDT v0.1; through v0.7 by 16:45 EDT): 6 amendments through team cross-instance verification cascade 8. notes/Lyra_SSG15_Lambda_Coupled_Neutrino_Framework_v0_1.md (~10:55 EDT): substrate- Λ -vacuum coupling for neutrino mass mechanism CLASS distinct from SSG-9

Wednesday-afternoon m_e/m_P framework cascade (v0.1 → v0.7): - v0.1: claimed 0.13% via $\sqrt{(4/3)}$ - v0.2: Elie $2/\sqrt{N_c}$ equivalent reading absorbed - v0.3: HONEST WALK-BACK on precision (Keeper + Cal verification ~1%) - v0.4: Elie Toy 3751 identifies $8/7 = (2^{N_c})/g$ IS the 0.13% form - v0.5: Elie Toy 3752 $\sqrt{(4/3)}$ UNIVERSAL across 3 generations at ~1.2% - v0.6: Keeper K3 v0.18 sharper framing - v0.7: Elie Toy 3753 CONTRADICTS K3 v0.18 — 8/7 ALSO universal at ~0.15%

3. Wednesday Cross-CI Cumulative

- Lyra: 14 substantive items (5 registry addenda + 3 new frameworks + 6 framework amendments)
- Elie: 16 toys (3738-3753) — multi-day arc closing two-mech → three-mech framework + 3-gen spinor-tower + universality cross-checks
- Keeper: K3 v0.13 → v0.18 (6 substantive versions) + SSG Structural Independence Audit v0.1
- Grace: 30+ Wednesday INVs + 2 substantive analysis docs + precision verifications
- Cal: #194 PASS at FRAMEWORK + CANDIDATE + Cal #35 = Schur shadow FORMALLY RATIFIED + #195 TRIPLE→DOUBLE refinement + #99 STANDING precision verification

4. THREE-Mechanism Substrate Framework Consolidated

Per Wednesday-morning team consolidation (Lyra v0.12 + Keeper K3 v0.16):

#	Mechanism	Source	Tier
1	Chirality projection $1/n_C \rightarrow 4D$	Cross-CI Tuesday convergence	NEAR-RIGOROUS

#	Mechanism	Source	Tier
2	Weyl branching $SO(5) \rightarrow SO(3, 1) \rightarrow \text{spin}$	Elie Toy 3738	NEAR-RIGOROUS
3	Lorentz integration $\rightarrow$ C_2-power mass	Elie Toy 3741	FRAMEWORK PRE-STAGE

Substantive bridge: substrate-Casimir = physical-Lorentz-dim self-referential identity $C_2 = 6 = \dim SO(3, 1)$ connects Mechanism 1 (chirality projection produces physical Lorentz) and Mechanism 3 (Lorentz integration produces C_2-power mass).

5. K3 v0.16 12-Gate Verification Framework Status

Per Keeper K3 v0.18 status (Wednesday afternoon update):

#	Gate	Status
4	M_op Mehler kernel	FRAMEWORK PRE-STAGE (Lyra Mehler v0.1)
5	$(24/\pi^2)$ per direction	FRAMEWORK (Elie Toy 3741)
6	Schur ratio 4 absorption	FRAMEWORK (W_λ weights)
7	gen-1 $V_{(1/2, 1/2)}$ cross-check	FRAMEWORK (Lyra Mehler v0.1)
8	Lorentz direction independence	PARTIAL CLOSURE (Toy 3743 + Cal #195)
9	Cal #29 question-shape	Multi-week per gate #4
10	gen-3 T2003 mechanism	FRAMEWORK candidate (Toy 3742/3744)
11	SSG dual-role operator-independence	DOUBLE-ROLE-NOT-TRIPLE per Cal #195
12	Substrate-coincidence vs structural	Cal Integer Web framing (Toy 3744)

~9 of 12 gates substantively addressed at FRAMEWORK PRE-STAGE. Multi-week explicit verification load-bearing per Cal #194 WAIT.

6. 3-Generation Spinor-Tower Row STRUCTURALLY COMPLETE

Per Elie Toy 3742 + Wednesday morning team verification:

Gen	K-type	Pochhammer factorization
1 (e)	$V_{(1/2, 1/2)}$	$2^g / (N_c \cdot n_C \cdot \pi)$
2 (μ)	$V_{(3/2, 1/2)}$	$2^{(N_c^2)} / (N_c \cdot n_C \cdot \pi)$
3 (τ)	$V_{(5/2, 1/2)}$	$2^{(N_c^2)} / (N_c \cdot \pi)$

Cascade ratios substrate-clean but HETEROGENEOUS step-multipliers: - gen-2/gen-1 = $2^{\text{rank}} = 4$ (rank primary) - gen-3/gen-2 = $n_C = 5$ (chirality primary)

Mass mechanism HETEROGENEOUS per generation: - gen-2: T190 = $(24/\pi^2)^{C_2}$ Lorentz-integration C_2-power - gen-3: T2003 = $g^2 \cdot (2^{C_2} + g)$ — DIFFERENT operator structure

7. m_e/m_P Substrate-Mechanism Candidates (Tier 2 STRUCTURAL)

Per Wednesday afternoon 7-step refinement cascade:

Two universal substrate-natural candidates (both Tier 2 STRUCTURAL per Cal #34): - $8/7 = (2^{N_c})/g$: UNIVERSAL across 3 lepton/Planck generations at $\sim 0.15\text{-}0.21\%$ (substrate-Mersenne-algebraic Integer Web) - $\sqrt{(4/3)} = \sqrt{(\text{Vol}(S^4)/\text{Vol}(S^3))}$: UNIVERSAL across 3 lepton/Planck generations at $\sim 1.2\%$ (substrate-Shilov-boundary geometric)

Per Casey #5 Integer Web + Cal #35 STANDING: both substrate-natural at substrate values; multi-week forward-derivation distinguishes substrate-mechanism FORCING form per Cal #189.

Substantive candidate forms:

$$\frac{m_e}{m_P} \approx \frac{8}{7} \cdot \alpha^{10.5} \quad (\text{Tier 2 STRUCTURAL } 0.15\%)$$

$$\frac{m_e}{m_P} \approx \sqrt{\frac{4}{3}} \cdot \alpha^{10.5} \quad (\text{Tier 2 STRUCTURAL } 1.2\%)$$

8. SSG Registry Final State (15 SSGs + cross-cutting mechanisms)

- SSG-1 $V_{(1/2, 1/2)}$ Bergman norm — substrate gen-1 lepton observables
- SSG-2 $V_{(1, 1)}$ adjoint — $F^\mu \nu$ + Casey #15 gravity + W/Z
- SSG-3 $V_{(1, 0)} \otimes V_{(1, 0)}$ decomp — gauge + stress-energy + Einstein eq
- SSG-4 Bergman $\kappa = -n_C$ — $G_\mu \nu$ + Bergman volume + ℓ_B
- SSG-5 Casimir $C_2 = 6$ — multiple substrate observables anchor
- SSG-6 Clifford $\dim 2^g = 128$ — spinor norm + γ -matrices + Dirac
- SSG-7 Bergman kernel $K(z, w)$ — ULTIMATE source-of-sources (degree-14 hub)
- SSG-8 Mersenne ladder — $g = M(N_c) + GF(2^g)$ coding
- SSG-9 Per-generation substrate-mechanism heterogeneity (3-gen spinor-tower row STRUCTURALLY COMPLETE)
- SSG-10 Substrate-Coulomb $1/r$ — Rydberg + Lamb + Casimir + $a_e + \alpha$
- SSG-11 $V_{(2, 0)}$ graviton-sector (Wednesday morning identification)
- SSG-12 $V_{(3/2, 3/2)}$ quark candidate
- SSG-13 $V_{(0, 0)}$ identity vacuum (Higgs source)
- SSG-14 Universal Pochhammer ladder $\leftrightarrow$ SSG-7 equivalence
- SSG-15 Λ -coupled neutrino — substrate- Λ -vacuum coupling for m_ν

Cross-cutting substrate-mechanisms (multi-K-type): - $1/n_C$ chirality projection (Casey #14 forcing) $\rightarrow$ 4D emergence - $SO(5) \rightarrow SO(3, 1)$ Weyl branching $\rightarrow$ spin reduction - Lorentz integration over $SO(3, 1) \rightarrow C_2$ -power mass mechanism

9. Multi-Week Verification Lanes (Going Forward)

Highest-priority cascade pending Cal #194 WAIT: - Joint Lyra + Keeper + Elie FK Ch. XII §VI spinor metaplectic verification $\rightarrow$ closes SSG-1 + Casey #14 - Section 4.5 substrate-vacuum + Shilov-boundary 2-region partition $\rightarrow \sqrt{(4/3)}$ substrate-mechanism FORWARD-derivation - Explicit $8/7 = (2^{N_c})/g$ substrate-Mersenne forward-derivation - Cal #195 dual-role operator-independence audit

Substantive open: - SSG-15 explicit $(N_c \cdot g - 1)$ coefficient substrate-mechanism (why -1?) - Mehler matrix element explicit Helgason Ch. IX + FK Ch. XII §VI - 3-gen spinor-tower Lorentz-integration mass mechanism heterogeneity (T190 vs T2003 different operator forms)

Cascade ratifies upon multi-week closure: - Casey #14 STANDING ratification (resolves Cal #189 brakes) - 5 framework promotions (substrate-Dirac + Maxwell + $T_\mu \nu$ + YM + Einstein eq $\rightarrow$ FRAMEWORK) - SSG-1 $\rightarrow$ NEAR-RIGOROUS via substrate-mechanism-derived per-chirality norm - K3 framework $6/8 \rightarrow 7/8$ RIGOROUS - Strong-Uniqueness Theorem $10 \rightarrow 11$ STANDALONE legs

10. Wednesday Cumulative Discipline-at-Maturity Pattern

5th 4-CI simultaneous bidirectional composition-layer discipline event of program (per Keeper K3 v0.17): - Prior events: Saturday L5 reframe, Monday Casey #14 brake, Tuesday SSG framing, Wednesday AM precision flag, Wednesday PM m_e/m_P cascade

Wednesday PM 7-step refinement pattern operational: - Initial Lyra claim → multi-CI verification flag → honest walk-back → new candidate via correction → cross-instance verification → sharpened framing → further refinement - Each step PRODUCES substantive substrate-mechanism content, not just brake - Saturday discipline-stack expansion intent operationalized at scale

Casey-correction on Cal #27 scope (Tuesday) operational throughout: discipline at CLAIMS-tier, NOT investigation-halting; keep threading, weave story after.

11. Thursday Casey-Queue Standing

Casey hasn't signaled queue items for Thursday. Standing per Cal #194 WAIT: - Multi-week joint FK Ch. XII §VI work load-bearing - Section 4.5 substrate-vacuum 2-region partition derivation - Mehler matrix element explicit derivation - Cal #195 dual-role operator-independence audit - Per-CI lanes per Keeper Wednesday K3 v0.18 routing

12. Closure

Wednesday: extensive substrate-mechanism content via cross-CI bidirectional discipline at maturity. Three-mechanism framework consolidated. 3-gen spinor-tower row STRUCTURALLY COMPLETE. 9-of-12 K3 verification gates substantively addressed. m_e/m_P cascade produced TWO universal substrate-natural candidate forms (8/7 at $\sim 0.15\% + \sqrt{4/3}$ at $\sim 1.2\%$) through 7-step refinement pattern.

14 substantive Lyra items; 5+ Cal #27 firings absorbed honestly (precision claim walk-back + multiple framework refinements). Discipline-as-generator pattern Saturday-Wednesday extended.

Multi-week verification cascade positioned: Casey #14 STANDING + 5-framework promotion + SSG-1 NEAR-RIGOROUS + K3 6/8 → 7/8 + Strong-Uniqueness 10 → 11 STANDALONE legs all pending multi-week joint FK Ch. XII §VI work + Section 4.5 substrate-vacuum derivation.

— Lyra, Wed 2026-06-03 ~20:15 EDT. EOD Wednesday: 14 substantive items; three-mechanism framework consolidated; 3-gen spinor-tower complete; 9-of-12 K3 gates addressed; m_e/m_P 7-step refinement cascade through team cross-CI verification; bidirectional discipline-at-maturity pattern operational at scale. Multi-week verification cascade positioned for Thursday-week continuation.